AI and the Career Ladder

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Corvinus Doctoral Seminar, 2026. 10. 14.

From an AI job apocalypse in 2025 to an AI jobs boom in 2026

AXIOS

Behind the Curtain: A white-collar bloodbath

Axios, May 2025

The New York Times

For Some Recent Graduates, the A.I. Job Apocalypse May Already Be Here

New York Times, May 2025

THE WALL STREET JOURNAL.

Companies Predict 2026 Will Be the Worst College Grad Job Market in Five Years

Wall Street Journal, November 2025

THE WALL STREET JOURNAL.

College Graduates Are Finally Catching a Break in This Job Market

Wall Street Journal, April 2026

The Economist

Is AI putting graduates out of work already?

The Economist, May 2026

The Economist

The jobs apocalypse is postponed. An AI jobs boom is here

The Economist, September 2026

AI and jobs: heavily debated, no consensus

- A new paper on AI and jobs almost every week
- US: employment of young workers in AI-exposed jobs fell 19%
- Europe: small or no effects
- AI exposure, adoption and their labor market effects can differ with a country's development and labor market tightness

What AI could do to the career ladder

- Gives beginners the know-how that used to take years of experience to learn
- Takes over the routine tasks that junior workers start with

Has AI reduced employment in exposed occupations, and has it changed the return to seniority?

- Hungarian administrative linked employer–employee data
- AI exposure linked at the occupation level
- Hungary: less developed than the US and Western Europe, with a tight labor market

Preview of findings

- No decline in employment; if anything, the most exposed occupations grew fastest
- Among young workers, employment grew most where AI augments work
- Young workers in exposed occupations gain about 4–6% in wages relative to the unexposed
- The gain fades with age $\rightarrow$ suggestive evidence of a decreasing return to seniority

1. AI and employment: evidence by country

		Data end	Junior (22-25)	Senior (50+)	Wages
US	Brynjolfsson et al. (2026)	Jun 2026	−19% negative	≈ 0 zero	≈ 0 zero
UK	Klein Teeselink (2025)	Jun 2025	−0.4% negative	≈ 0 zero	−1.1% negative
Sweden	Lodefalk et al. (2026)	Jun 2025	−5.5% negative	+1.3% positive	–
Denmark	Humlum & Vestergaard (2026)	Dec 2024	≈ 0 zero	≈ 0 zero	≈ 0 zero
Finland	Kauhanen & Rouvinen (2026)	Sep 2025	zero/positive	≈ 0 zero	≈ 0 zero
Norway	Hernæs & Kostøl (2026)	Feb 2026	≈ 0 zero	–	–

Most vs. least AI-exposed occupations. UK: posted salaries in job ads.

2. Expertise

- AI raises productivity most for less experienced and less skilled workers, and the young adopt it first (Noy & Zhang, 2023; Brynjolfsson, Li & Raymond, 2025; Humlum & Vestergaard, 2025)
- Expertise loses value: automating expert tasks lowers wages (Autor & Thompson, 2025); where job skills change fast, wages rise less with experience (Deming & Noray, 2020)

3. Automation in the US and Europe

- Robots cut jobs in the US, not in Germany (Acemoglu & Restrepo, 2020; Dauth et al., 2021)
- Scarce labor speeds up automation (Acemoglu & Restrepo, 2022)

- **Linked employer–employee administrative panel**
  - worker × firm × month, January 2021 – December 2025
  - earnings, employer, occupation (4-digit FEOR-08)
- **AI exposure** by 4-digit FEOR-08 crosswalked from SOC codes

Sample

- positive earnings
- full-time workers (at least 40 hours a week)
- age: 22-65
- exclude: armed forces and law enforcement, health-care occupations

Exposure: what an LLM could do, and what people use it for

Both measures are built on **O*NET tasks**: the list of tasks the US O*NET database records for each occupation ▸ Example: economists' tasks

- β **index**: what an LLM could do hypothetically (Eloundou et al., 2024)
  - The share of an occupation's O*NET tasks an LLM could do in half the time, rated by GPT-4
- **Anthropic Economic Index (AEI) usage**: what people actually use AI for (Handa et al., 2025)
  - The share of all Claude conversations that can be linked to the O*NET tasks of an occupation
  - **Automation and augmentation (AEI)**: does the user delegate the task to AI, or collaborate with it? ▸ Automation and augmentation: examples

We rank occupations by each measure, split the ranking into quintiles

▸ Most and least exposed occupations ▸ Most and least exposed: automation and augmentation

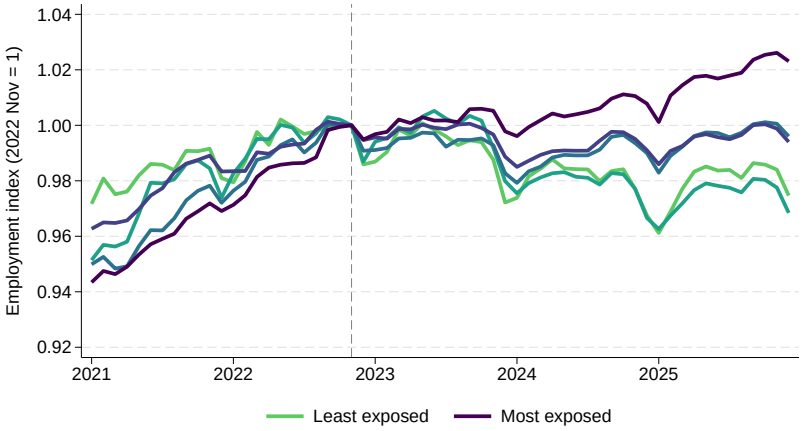
O*NET tasks → O*NET-SOC → US SOC 2010 → ISCO-08 → **FEOR-08** (4-digit)

- **Starting point:** β and usage are measured for US occupations (O*NET-SOC), then aggregated to US SOC 2010 (6-digit)
- **SOC → ISCO-08:** official BLS crosswalk
- **ISCO-08 → FEOR-08:** official KSH crosswalk
- **Coverage**: number of occupations

	All	With β	With AEI usage
US SOC 2010	840	778	587
FEOR-08	485	483	428

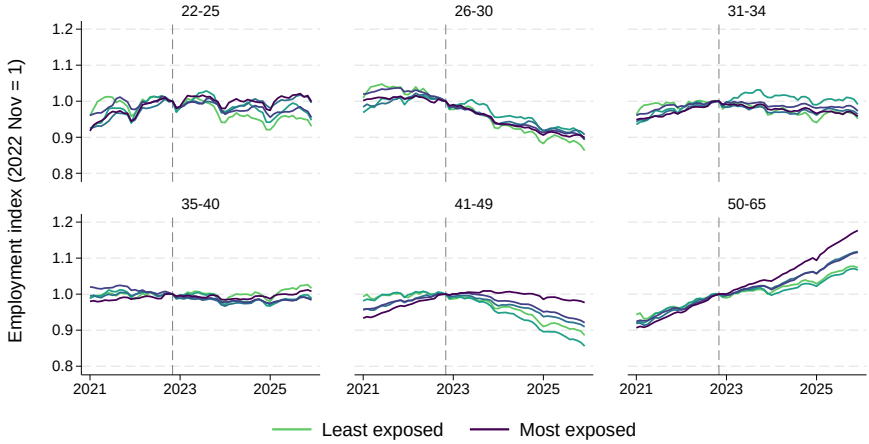
▸ Example: economic analyst

Employment by AI exposure quintile: AEI usage



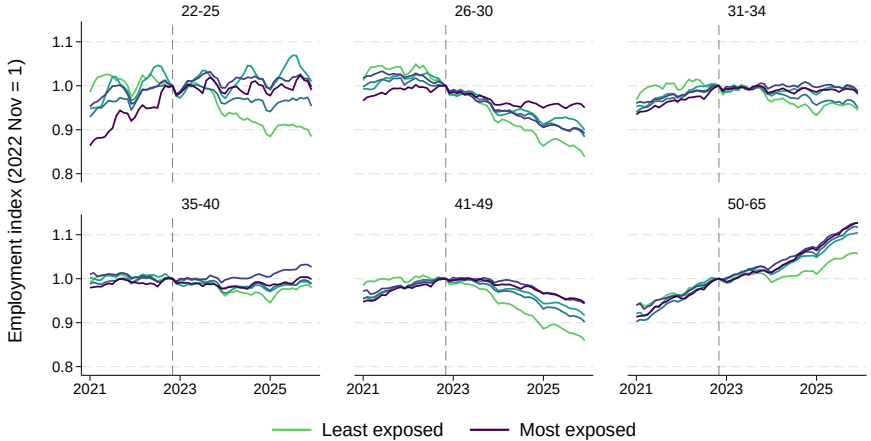
▸ β quintile

Employment by age and exposure: AEI usage

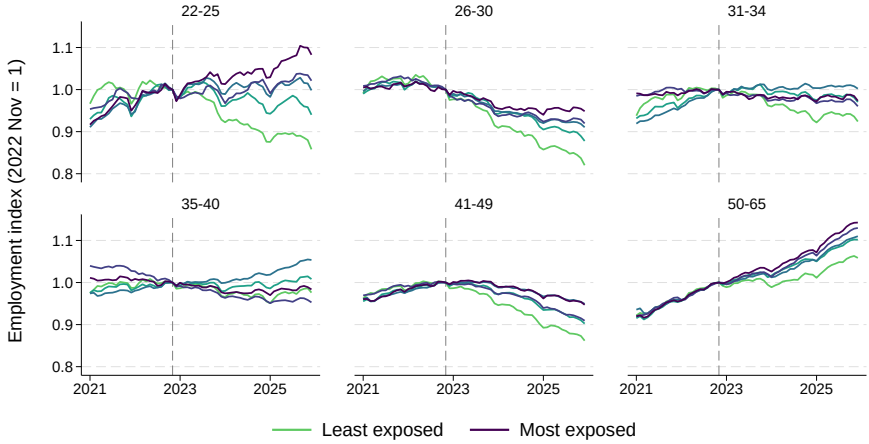


▸ β quintile ▸ Cohort demographics

Automation


22-25
26-30
31-34
35-40
41-49
50-65
Employment index (2022 Nov = 1)
1.1
1.0
0.9
0.8
2021
2023
2025
Least exposed
Most exposed

Augmentation



$$\ln w_{it} = \delta \, \text{Post}_t \times \text{Exp}_{it} + \sum\nolimits_a \theta_a \, \text{Post}_t \times \text{Exp}_{it} \times \text{Age}_{it}^a + \gamma X_{it}' + \alpha_i + \lambda_t + \varepsilon_{it}$$

w_{it}: gross monthly earnings of worker i in year t
$\text{Post}_t = 1$ from 2023 on
$\text{Exp}_{it} = 1$ if in the top two exposure quintiles
$\text{Age}_{it}^a = 1$ if in age group a: 22–25 (baseline), 26–30, 31–34, 35–40, 41–49, 50–65
X_{it}: other main effects and two-way interactions
α_i, λ_t: worker and year fixed effects

Sample: only November of every year

- δ: extra wage growth after 2022, exposed vs. unexposed, ages 22-25
- θ_a: difference for other age groups, compared to the baseline (22-25)

Change after 2022 in the wages of exposed vs. unexposed workers

Age	Exposed by β	Exposed by AEI usage
22–25 (δ)	0.055***	0.042***
	(0.012)	(0.015)
Older ages, difference from 22–25 (θ_a)		
26–30	−0.023***	−0.006
	(0.007)	(0.007)
31–34	−0.051***	−0.022**
	(0.009)	(0.010)
35–40	−0.082***	−0.034**
	(0.014)	(0.016)
41–49	−0.100***	−0.050***
	(0.014)	(0.017)
50–65	−0.093***	−0.048***
	(0.014)	(0.018)
Observations	18,457,338	18,471,790
R^2	0.775	0.775

SEs clustered by occupation. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

▸ $\delta + \theta_a$

In a less developed economy with a tight labor market:
no job losses from AI, but a smaller return to seniority

- **Employment:** no decline in exposed occupations
- **Young workers:** employment grew most where AI augments work
- **Wages:** the gain of exposed young workers fades with age, return to seniority decreases

Most and least exposed occupations: β and AEI usage

β: share of tasks an LLM could do	
Most exposed	
2142	Software developers
3641	Personal assistants
3614	Accounting clerks
3611	Financial clerks
4190	Other office clerks
Least exposed	
5243	Building caretakers
9235	Fast-food sellers
9329	Simple construction workers
8422	Earthmoving machine operators
9236	Kitchen helpers

AEI usage: share of conversations	
Most exposed	
2142	Software developers
2159	Database and network analysts
2149	Other software developers
2410	University teachers
2141	Systems analysts
Least exposed	
8219	Product assemblers
7511	Bricklayers
8421	Agricultural machine operators
9225	Hand packers
5243	Building caretakers

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Automation and augmentation: does the user delegate, or collaborate?

Each Claude conversation is one type; an occupation's score is the **share of its conversations** of each type

- **Automation**: AI executes the task with minimal human involvement
  - directive: *"Format this technical documentation in Markdown"*
  - feedback loop: *"Here's my Python script for data analysis – it's giving an IndexError. Can you help fix it? ... Now I'm getting a different error..."*
- **Augmentation**: the user collaborates with AI
  - task iteration: *"Let's draft a marketing strategy for our new product. ... Good start, but can we add some concrete metrics?"*
  - learning: *"Can you explain how neural networks work?"*
  - validation: *"I've written this SQL query to find duplicate customer records. Can you check if my logic is correct and suggest any improvements?"*

Examples from Handa et al. (2025)

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Most and least exposed occupations: automation and augmentation

Automation: AI does the task	
Most exposed	
3641	Personal assistants
2142	Software developers
2152	System administrators
4224	Customer information clerks
4222	Receptionists
Least exposed	
2312	Social workers
2212	Medical specialists
9331	Simple agricultural workers
5117	Cashiers
3623	Purchasing clerks

Augmentation: AI helps the worker	
Most exposed	
3612	Bank clerks
2910	Other highly qualified administrators
2141	Systems analysts
2410	University teachers
1210	Chief executives
Least exposed	
5134	Cooks
7223	Cabinetmakers
3623	Purchasing clerks
4112	Office administrators
5299	Other service workers

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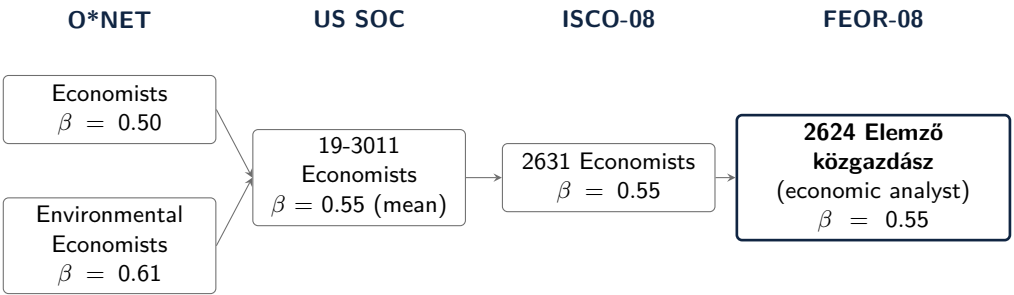
Economists' O*NET tasks

O*NET task	% of all conversations	Automation (%)	Augmentation (%)
Compile, analyze and report data; forecast market trends	0.058	26	70
Advise businesses and agencies on economic relationships	0.057	16	76
Formulate recommendations or policies for economic problems	0.027	19	76
Conduct research; publish reports or journal articles	0.011	0	79
Study economic and statistical data in a specialization	0.009	17	69
...	...	...	...

O*NET occupation 19-3011.00 Economists. AEI, March 2025 release. Automation and augmentation are shares of the task's conversations; the rest is unclassified.

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Crosswalk example: Elemző közgazdász, i.e. economic analyst (FEOR 2624)



β is a share of tasks, so the SOC code takes the mean of its two O*NET occupations. Economic analysts get all their β from ISCO 2631, so it carries over unchanged.

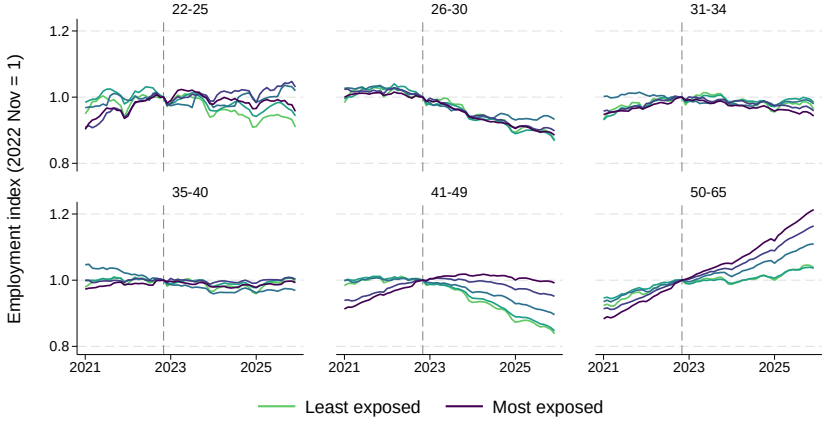
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Employment by β quintile



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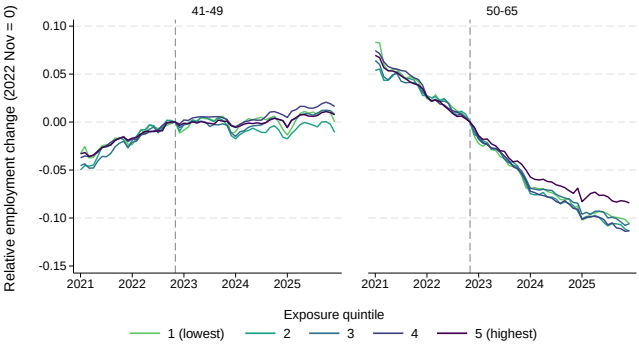
Employment by age and β quintile



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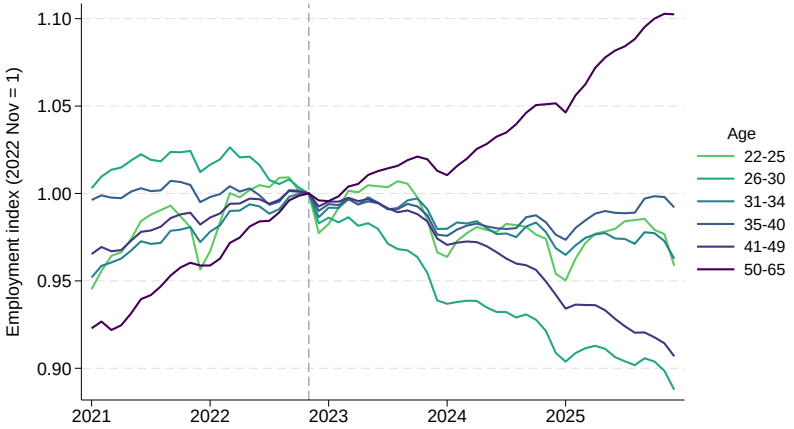
Ages 41–65: the gap between quintiles is cohort composition, not AI

- Ratkó cohorts age into 50+; entrants are more educated (1990s college expansion)
- Change of each birth cohort since November 2022 (= 0, not 1): the gap mostly disappears
- 50–65 grows in the main figure as large cohorts age in, but falls here as each cohort retires

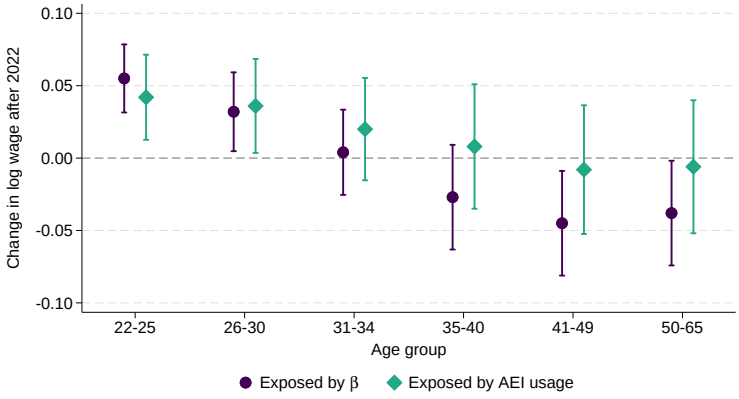


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Employment by age group



Change in wages after 2022 by age group, exposed vs. unexposed: $\delta + \theta_a$



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